Ex No 5 2219126

Practical - 5

AIM:

To Expiment on Packet capture tool : Wiresharк

Packet Sniffer:

- Sniffs messages being sent/recieved from/by your computer
- Store and dyplay the contents of the various protocol fields in the messages
- Passive Program
  - never sends packets itself
  - no packets addressed to it
  - recieves a copy of all packets

Packet Sniffer Structure Diagnostic Tools

- Tcdpump
  - Eg. tcdump -enx host 10.129.41.2 -w ex.out
- wireshark
  - wireshark -r ex3.out

Description:

wireshark:

universal

Wireshark a network analysis tool formerly known as Ethereal, captures packets in real time and dyplay them in human readable format. Wireshark includes filters, color coding, and other features that let you dig deep into network traffic and inspect individual packets. You can use Wireshark to inspect a suspicious program's network traffic, analyze the traffic flow on your network, or troubleshoot network problems

What can we do with wireshark

- Capture network traffic
- Decode packet protocols using dissectors
- Define filters - capture and display
- Watch smart statistics
- Analyze Problems
- Interactively browse that traffic

Wireshark used for:

- Network administrators : troubleshoot network problems
- Network security engineers : examine security problems
- Developers : debug protocol implementations
- People : learn network protocol internals

Capturing and analysing packets using wireshark tool

To filter, capture, view packets in Wireshark Tool.

Capture 100 packets from the Ethernet. IEEE 802.3 LAN Interface and save it.

Procedure:
- Select LAN in wireshark
- Go to capture option
- Select stop capture automatically after 100 packets
- Then click start capture
- Save the packets

1. Create a Filter to display only TCP/UDP packets, inspect the packets and provide the flow graph

Procedure:
- Select LAN in Wireshark
- Go to capture option
- Select stop capture automatically after 100 packets
- Then click start capture
- Search TCP Packets in search bar.
- To see flow graph click Statistics → Flow graph
- Save the packets.

2. Create a filter to display only ARP packets and inspect the packets

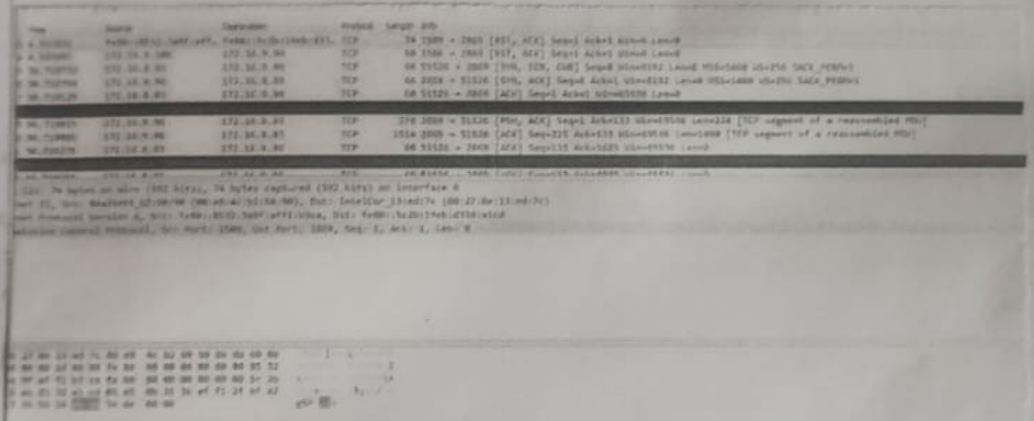
Procedure:
- Go to capture option
- Select stop capture automatically after 100 packets
- Then click start capture
- Search ARP packets in search bar
- Save the packets

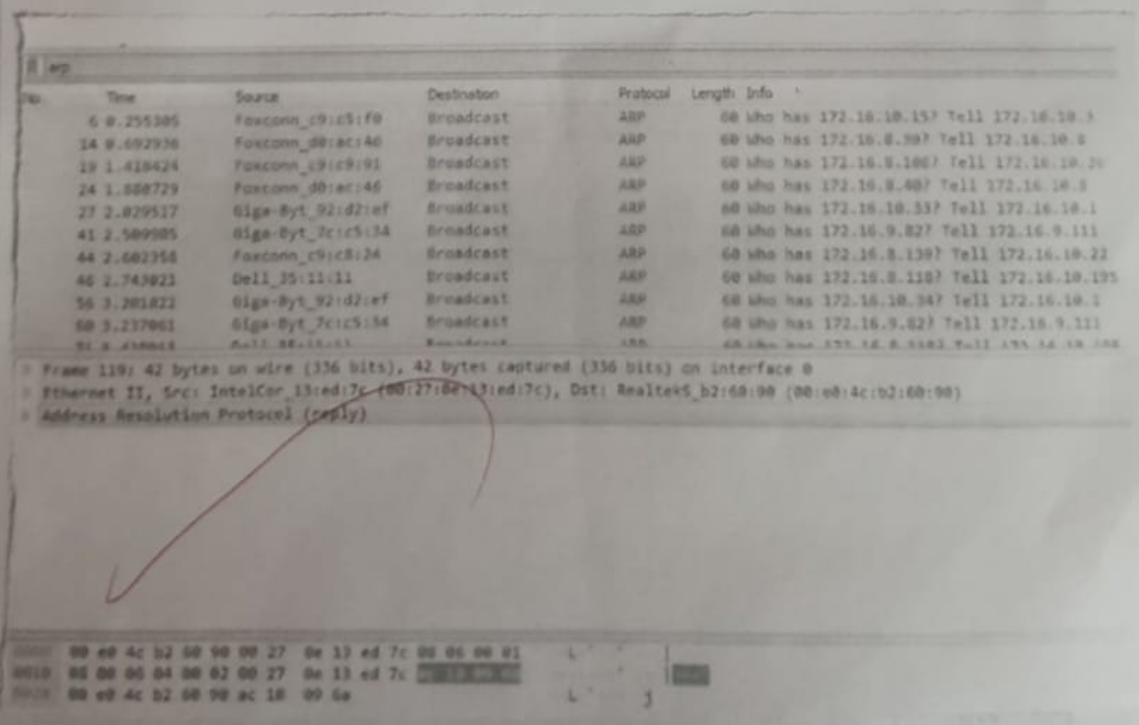
3. Create a filter to display only DNS packets and provide the flow graph. Procedure
- Go to capture option
- Select stop capture automatically after 100 packets
- Then click start capture
- Search DNS packet in search bar
- To see flow graph click Statistics → Flow graph
- Save the packets

OUTPUT

1. Create a filter only to display TCP Packets, inspect the packets

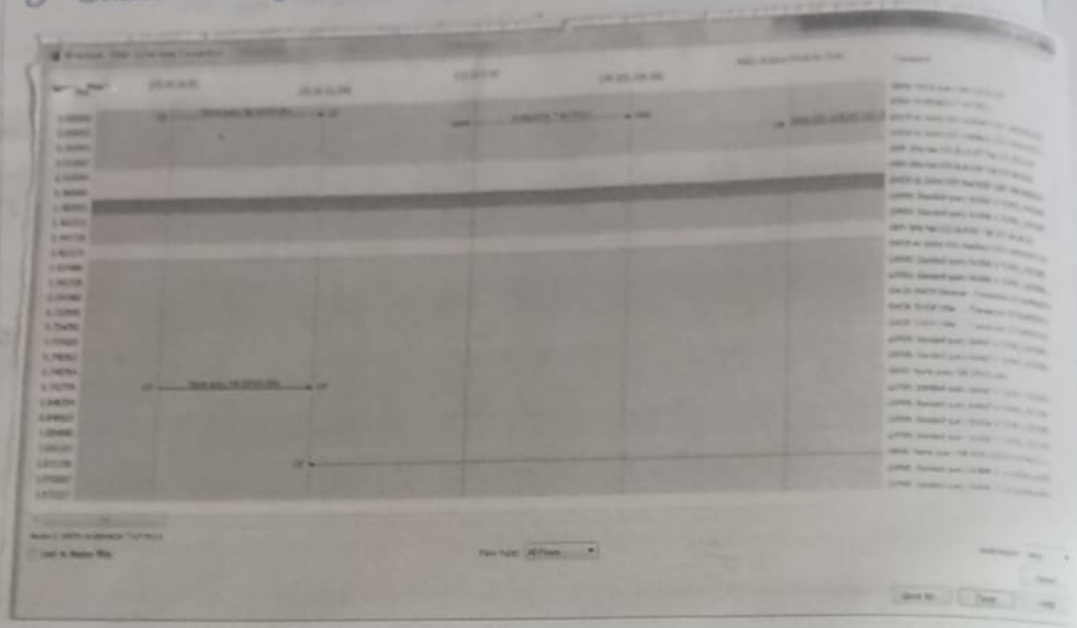
2. Create a filter only to display ARP Packets





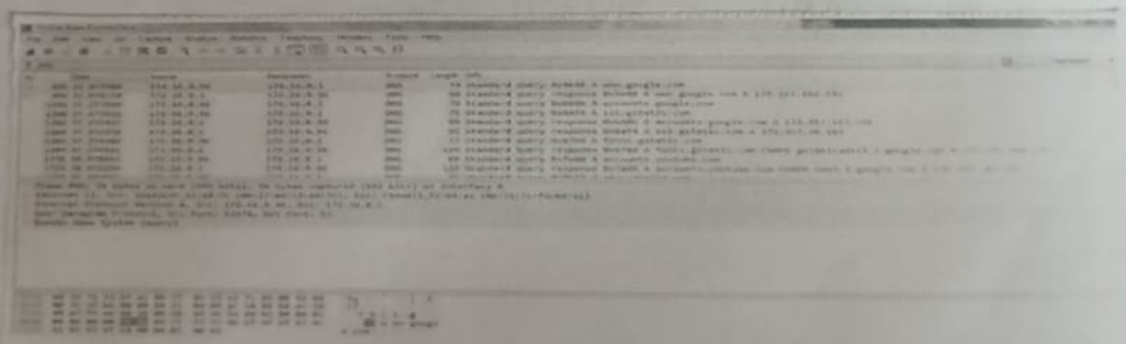
Ex no 6

3. Create a filter only to display DNS packets



4 Create a filter only to display DHCP Packets

- Select LAN connections in wireshark . Go to capture option
- Select stop capture automatically after 100 packets . Then click start capture
- Search DHCP packets in search bar . Save the packets



Result:

The experiment on wireshark is completed successfully